

Artificial Intelligence for Digital Characters

Solution Exercise 3

Hand-out Date: 22 April 2026



Goals

- Understand the key concepts in speech synthesis, including text normalization, prosody, prominence, and different synthesis techniques.
- Understand the architecture and functionality of acoustic models in text-to-speech (TTS) systems, including Tacotron 2 and FastSpeech.
- Explore vocoder techniques for waveform generation, comparing classical signal processing methods and modern neural vocoders such as WaveNet, HiFiGAN, and DiffWave.
- Analyze methods for controlling speaker identity and speaking style in TTS using one-hot labels, learned embeddings, and style reference encoders.
- Develop an understanding of zero-shot and few-shot adaptation techniques for speaker/style transfer in speech synthesis.

Resources

The lecture slides, exercise slides, and Unity template are accessible via Moodle.

Tasks

1. Multiple Choice Questions

Please mark the correct answer for the multiple-choice questions below.

- a) False – It also includes things like number expansion, disambiguating punctuation, etc.
- b) True – Machine learning models can be trained to detect sentence boundaries using context and punctuation cues.
- c) False – Pitch (F0), loudness, and duration all play roles in prosody.
- d) False – It involves pitch, duration, and sometimes loudness — it’s about emphasis.
- e) False – Lapse is when they are too far apart
- f) True – Diphone synthesis often lacks natural prosody because it uses a limited set of prerecorded units.
- g) True – Unit selection synthesis produces more natural speech by choosing larger, more contextually appropriate segments.
- h) False – They use pre-recorded segments of real speech.
- i) False – The question describes an insertion error, not a deletion. A deletion occurs when the system (SYS) completely misses a word that was present in the correct transcript (REF).
- j) True – MOS uses an absolute scale (1-5 averaged), while A/B tests ask listeners for a comparative preference (“prefer A, prefer B”).
- k) False – words in a Diagnostic Rhyme Test (DRT) are “Often embedded in carrier phrases” (e.g., “now we will say <word> again”), rather than being played strictly in isolation.

2. Acoustic Models

a) Tacotron 2:

1. Uses: Autoregressive decoder + attention mechanism. (implicit duration modelling via attention)
2.
 - Advantage: Generates high-quality, natural-sounding speech due to flexible alignment via attention.
 - Limitation: Slow inference due to sequential decoding (step-by-step mel prediction).

FastSpeech:

1. Uses: Non-autoregressive decoder + duration predictor.
2.
 - Advantage: Fast and parallel inference due to pre-determined durations and full-sequence decoding.
 - Limitation: Relies on external duration annotations or aligners during training; can reduce prosody flexibility.

b)

1. Attention in Tacotron 2:

- Computes alignment between encoder outputs h_j and decoder state s_{i-1} .
- Uses location-sensitive attention with convolution over previous alignments (adds temporal structure).
- Produces context vector $c_i = \sum_j \alpha_{ij} h_j$, where α_{ij} is a soft attention score.

2. FastSpeech motivation:

- Tacotron 2's attention is unstable (can skip/repeat words) and slow due to autoregressive generation.
- FastSpeech uses duration prediction + length regulation to remove attention, allowing full-sequence parallel decoding.

3. Benefit of FastSpeech:

- Better for real-time or low-latency applications.
- More stable and deterministic output.

c)

1. Step 1: Upsampling by repeating hidden states
 - h_1 repeated 2 times, h_2 repeated 1 times, h_3 repeated 3 times.
 - Resulting sequence: [[0.1, 0.2], [0.1, 0.2], [0.3, 0.1], [0.2, 0.4], [0.2, 0.4], [0.2, 0.4]]
2. Step 2: Efficiency gain explanation:
 - Since durations are known in advance, the decoder doesn't wait for previous outputs (non-autoregressive).
 - Enables parallel computation and real-time synthesis, unlike Tacotron 2 which must decode frame by frame.

d)

1. Two challenges:
 - The word “present” appears twice with different pronunciations (verb vs noun).
 - Requires semantic understanding and correct stress (e.g., preSENT vs PREsent).
2. Resolve ambiguity in pronunciation and prosody:
 - Fine-grained style embeddings can guide prosody, rhythm, and intonation.
 - Embeddings can capture patterns from training data: e.g., formal tone, interrogative vs declarative delivery.
 - Speaker embeddings help maintain consistent accent or emotion.
3. Features that would help:
 - Style control token like “formal” or “neutral” could guide tone.
 - Speaker ID could help with timbre/accent that fits sentence context (e.g., news anchor vs storyteller).

3. Vocoders

a) Classical vs Neural Vocoders: Compare Griffin-Lim and WaveNet vocoders in terms of methodology and performance.

1. Input features:

- Griffin-Lim: Magnitude spectrogram
- WaveNet: Mel-spectrogram or linguistic features

2. Strength and limitation:

- Griffin-Lim:
Strength: Simple, no training required
Limitation: Noisy, robotic output
- WaveNet:
Strength: Very high-quality output
Limitation: Slow inference (sample-by-sample generation)

3. Lightweight and fast, making it ideal for prototyping, testing, or low-resource environments (e.g., embedded devices, real-time debugging).

b)

1. The roles of the generator, multi-scale discriminator, and multiperiod discriminator:

- Generator: Converts mel-spectrogram into time-domain waveform using upsampling and residual blocks.
- Multi-Scale Discriminator: Examines outputs at multiple resolutions to detect artifacts at different time scales.
- Multi-Period Discriminator: Focuses on periodicity (e.g., pitch cycles) to ensure natural prosody.

2. How the loss functions (real/fake, feature loss, L1) encourage the generation of realistic and natural-sounding speech:

- Real/Fake Loss: Discriminator distinguishes real from fake; generator tries to fool it (adversarial training).
- Feature Loss: Minimizes distance between intermediate activations for real and generated waveforms; improves perceptual quality.
- L1 Loss: Ensures output waveform's mel-spectrogram closely matches that of real audio (stabilizes training).

3. Advantages of HiFiGAN over WaveNet:

- HiFiGAN is non-autoregressive and fully parallelizable, resulting in real-time inference.
- WaveNet has higher quality in theory, but HiFiGAN achieves comparable quality at much lower cost.

c)

1. Transposed convolutions:

- Increase the temporal resolution of feature maps.
- Allow generator to go from compressed mel features (low resolution) to a full-resolution audio signal.

2. Parallel residual blocks with different dilation rates:

- Allow the model to capture local and global context.
- Dilation controls how much of the input history the network can “see.” Combining blocks with various dilation rates ensures the network captures both fine and coarse details (e.g., phoneme structure and prosody).

d)

1. Forward and reverse processes in the DiffWave:

- Forward process: Adds Gaussian noise over multiple steps to audio → creates a noisy version.
- Reverse process: Learns to remove the noise step-by-step, conditioned on the mel-spectrogram.
- The model is trained to predict the noise added at each step (MSE loss between predicted and true noise).

2. Trade-off between generation quality and inference time:

- DiffWave produces very high-quality audio (competitive with WaveNet and HiFiGAN).
- But it's slow at inference because it requires many iterative denoising steps.
- HiFiGAN is much faster and more suitable for real-time synthesis, even if DiffWave has slightly better fidelity.

3. Applications:

- Studio-quality voice cloning.
- Audiobook narration
- Post-processing or offline batch TTS where latency is not critical but naturalness and detail are prioritized.

4. Style Transfer

a)

1. One-Hot Labels vs. Learned Embeddings in TTS:

- One-Hot Labels: Represent speakers/styles using binary vectors where only one element is '1' and the rest are '0'. Each speaker/style has a unique, non-learned vector.
- Learned Embeddings: Use trainable dense vectors that are learned during model training to represent speakers/styles.

2. Advantage and limitation:

- One-Hot Labels:
 - Advantage: Simple to implement and train; easy to interpret
 - Limitation: Does not scale well (requires a new label for each speaker/style)
 - Inference: Needs known label at inference; cannot generalize to unseen speakers/styles.
- Learned Embeddings:
 - Advantage: Captures complex relationships in speaker/style space
 - Limitation: Requires large and diverse data for good generalization
 - Inference: Enables zero-shot or few-shot synthesis if trained well

b)

1. Pretrained on a separate speaker classification task (e.g., using CNN/RNN).

- At training time: embeddings are fed into the TTS model alongside the text input.
- At inference time: new speaker audio is passed through the embedding extractor to generate a representation.

2. Zero-shot synthesis is possible if:

- The training dataset is diverse enough across speakers.
- The embedding extractor has learned generalizable speaker characteristics (not overfitting).
- The TTS model is robust to variation in embedding space.

3. Benefits of pretrained/foundation models:

- Reduce need for large speaker-specific data.
- Improve generalization across accents, languages, and pitch ranges.
- Enable quick adaptation to new speakers with one or a few samples.

c)

1. Reference mel spectrogram:

- Fed into a style encoder during inference.
- The output (style embedding) captures prosody, pitch, rhythm, or emotion of the reference.
- This embedding is conditioned into the decoder to guide synthesis style.

2. Benefits of unsupervised style learning:

- No need for manual labeling, which is time-consuming and error-prone.
- Captures subtle prosodic features (e.g., emphasis, intonation) that are hard to define or annotate.
- More data-efficient for rare or mixed style attributes.

3. Real-world application example:

- Voice assistants that adapt tone based on context (e.g., “happy” for positive reminders, “serious” for alerts).
- Audiobook narration with emotion/character variation.
- Customer support bots that sound empathetic or formal depending on scenario.

d)

1. Learned embeddings are more appropriate for both:

- Speaker adaptation with limited data (supports few-shot/zero-shot).
- Emotion/style control using latent prosody representations.

2. Strategies:

- Use a reference encoder for style — accepts mel spectrogram of desired emotion.

- Use a speaker encoder pretrained on speaker ID task; freeze during TTS training.
- Add multi-task loss: e.g., TTS loss + auxiliary classification for emotion/speaker ID.
- Use VAE-style latent space for unsupervised style disentanglement.
- Optionally: fine-tune on small datasets for new speakers or styles using PEFT or adapter layers.